


Rings

$$\underline{\mathbb{Q}, \mathbb{R}, \mathbb{C}} \quad \mathbb{Z}_p$$

$$\underline{\mathbb{Z}, M_n(\mathbb{R})} = \{ n \times n \text{ matrices w/ real entries} \}$$

Def A ring is a set R with two binary operations $+$ and $\cdot$ satisfying the follow:

1) $(R, +)$ is an abelian group

2) $\cdot$ is associative: $a(b \cdot c) = (a \cdot b) \cdot c$

$\nexists e$ such that $e \cdot a = a = a \cdot e$

3) Distributive law $a \cdot (b + c) = a \cdot b + a \cdot c$
 $(b + c) \cdot a = b \cdot a + c \cdot a$

Def a ring $(R, +, \cdot)$ is called commutative
if $\cdot$ is commutative.

If $\cdot$ is not commutative a ring is
called a non-commutative ring.

Eg

1) Every field is a commutative ring.

2) $\mathbb{Z}$ is a ring

3) $M_n(\mathbb{R})$, $M_n(\mathbb{Q})$, $M_n(\mathbb{Z}_p)$

4) $\mathbb{Z}_n$ is a ring ($\mathbb{Z}_p$ is a field)

5) $F(\mathbb{R}, \mathbb{R}) = \{f: \mathbb{R} \rightarrow \mathbb{R}\}$ a ring under

$$(f+g)(x) = f(x) + g(x)$$

$$(f \cdot g)(x) = f(x) \cdot g(x)$$

Note under
functional composition
this is not necessarily

$$c) \mathbb{Z}[x] = \{\text{polynomials with coefficients in } \mathbb{Z}\}$$

$p(x) + q(x)$ is normal polynomial addition

$p(x) \cdot q(x)$ is normal polynomial multiplication

Look at $\mathbb{Z}_n$ when $n=6$

$\cdot$	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

$$\mathbb{Z}_6^* = \{1, 5\}$$

in $\mathbb{Z}, \mathbb{R}, \mathbb{Q}, \mathbb{C}$

if $a \cdot b = 0$ then $a = 0$ or $b = 0$

in $\mathbb{Z}_6$

$$2 \cdot 3 = 0 \quad \& \quad 2 \neq 0 \quad \& \quad 3 \neq 0$$

"zero divisor"

$$5^{-1} = 5$$

$$1^{-1} = 1$$

these are the only elements of $\mathbb{Z}_6$ with mult. inverses

$M_n(\mathbb{R})$ real square matrices

additive identity $\begin{pmatrix} 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{pmatrix}$ zero matrix

multiplicative identity $\begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}$ identity matrix

in general $AB \neq BA$ so this is an example of non-commutative ring.

"zero divisors" exist in $M_n(\mathbb{R})$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$M_n(\mathbb{R})^* = GL_n(\mathbb{R})$$

some matrices are invertible, & some are not.

$$GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det(A) \neq 0\}$$

call the units in a ring

Def Let R be a ring, then $R^* = \{x \in R : xy = 1 = yx \text{ for some } y \in R\}$ is called set of units.

Claim $(R^*, \cdot)$ is a group

Ps $\cdot$ is associative by definition of ring

Let 1 be the multiplicative identity

identity

$$1 \cdot 1 = 1 \quad \text{so} \quad 1^{-1} = 1 \Rightarrow 1 \in R^*$$

closed
under
mult

Let $x, y \in R^*$

$$xy y^{-1} x^{-1} = x 1 x^{-1} = x \cdot x^{-1} = 1$$

so $y^{-1} x^{-1}$ is a right inverse of xy

$$y^{-1} x^{-1} xy = 1 \quad \text{so} \quad y^{-1} x^{-1} \text{ is left inverse}$$

Aside

$$\boxed{\text{so } (xy)^{-1} = y^{-1} x^{-1}} \quad \text{so } xy \in R^*$$

inverse Let $x \in R^*$, by definition of R^*
 there exist y such that $xy = 1 = yx$.
 so $x^{-1} = y$. $\square$

Def A subring of a ring R is a subset of R
 that itself is a ring under the multiplication
 & addition of R .

Eg) $\mathbb{Z}$ is a subring of $\mathbb{Q}$, $\mathbb{Q}$ is a subring of $\mathbb{R}$

$M_n(\mathbb{Q})$ is a subring of $M_n(\mathbb{R})$ $\mathbb{Z}$

$\mathbb{Z}$ is a subring of $\mathbb{Z}[x]$ $\{0, 1, 3, 2, 4, 5\}$

non example $\mathbb{Z}_n$ is not a subring of $\mathbb{Z}$ $3 \cdot 5 = 15$

Prop 17.13 (Test for subrings)

Let $S \subseteq R$ a ring. S is a subring if and only if the following conditions hold:

1) $(S, +) < (R, +)$

2) $1 \in S$ 1 is multiplicative id.

3) closure if $a, b \in S$ then $a \cdot b \in S$

Prop 17.7 0 , additive inverse of R , is the unique element such that $0 \cdot a = 0 = a \cdot 0$.

PT $0 \cdot a = (0+0) \cdot a = 0 \cdot a + 0 \cdot a$

$$0 \cdot a = 0 \cdot a + 0 \cdot a$$

$$-0 \cdot a + 0 \cdot a = -0 \cdot a + 0 \cdot a + 0 \cdot a$$

$$0 = 0 + 0 \cdot a$$

$$0 = 0 \cdot a$$

a similar argument shows $a \cdot 0 = 0$

uniqueness suppose $0'$ has property $a \cdot 0' = 0' = 0' \cdot a$
 $0 = 0 \cdot 0' = 0'$ so $0 = 0'$.

Prop 17.8

Let R be a ring then for any $a, b \in R$

$$1) (-a) \cdot b = -(a \cdot b) = a \cdot (-b)$$

$$2) a \cdot (-b) = -a \cdot b$$

$$3) (-1) \cdot a = -a = a \cdot (-1)$$

Pf 1) $a \cdot (-b) + a \cdot b = a \cdot (-b + b) = a \cdot 0 = 0$

$$\text{So } a \cdot (-b) = -(a \cdot b)$$

$$(-a) \cdot b + a \cdot b = (-a + a) \cdot b = 0 \cdot b = 0$$

$$\text{So } (-a) \cdot b = -(a \cdot b)$$

$$\begin{aligned}
 2) \quad (-a) \cdot (-b) &= -((-a) \cdot b) & (-a) \cdot b &= -(a \cdot b) = a(-b) \\
 &= -(- (a \cdot b)) \\
 &= ab
 \end{aligned}$$

$$3) \quad (-1) \cdot a = -(1 \cdot a) = -(a) = -a \quad \square$$